

File Integrity Monitoring (FIM)

File Integrity Monitoring (FIM) is a security capability that monitors files and directories for unauthorized changes. It detects events such as file creation, modification, deletion, and permission changes, helping security teams identify suspicious activity and maintain system integrity.

Why is FIM Important?

- Detects unauthorized changes to critical files.
- Identifies potential malware or ransomware activity.
- Protects important system and application files.
- Supports compliance with standards such as PCI DSS, HIPAA, ISO 27001, and NIST.
- Enables faster incident detection and response.

Common Files and Directories to Monitor

- System configuration files
- Application configuration files
- Authentication and security-related files
- Log files
- Web server files
- Executable files and scripts
- Sensitive business data

Why Organizations Use FIM

Organizations implement File Integrity Monitoring to detect unauthorized file changes, improve security monitoring, maintain system integrity, support incident response, and meet regulatory compliance requirements.

Now, let's move on to the practical implementation of File Integrity Monitoring (FIM) using Wazuh. 👉

◆ Goal

Detect when a file is:

- Created
- Modified
- Deleted

◆ Step 1: Configure FIM on the Agent

👉 On Windows Agent

1. Go to:

C:\Program Files (x86)\ossec-agent\ossec.conf

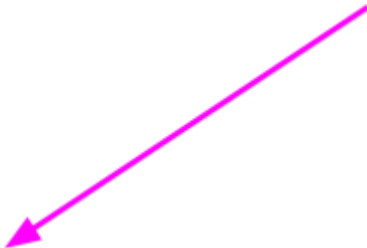
2. Add/modify this section:

- a. Create a new folder on Desktop and add the folder location **here**

```
<syscheck>
  <disabled>no</disabled>

  <!-- Monitor this folder -->
  <directories
check_all="yes">C:\Users\<username>\Desktop\Wazuh_FIM_Monitor</direc
tories>

  <frequency>60</frequency>
</syscheck>
```



```
ossec - Notepad
File Edit Format View Help
<disabled>no</disabled>
<windows_apps>./shared/win_applications_rcl.txt</windows_apps>
<windows_malware>./shared/win_malware_rcl.txt</windows_malware>
</rootcheck>

<!-- Security Configuration Assessment -->
<sca>
  <enabled>yes</enabled>
  <scan_on_start>yes</scan_on_start>
  <interval>12h</interval>
  <skip_nfs>yes</skip_nfs>
</sca>

<!-- File integrity monitoring -->
<syscheck>

  <disabled>no</disabled>

  <!-- I have add this manually to test and demo the FIM -->
  <directories check_all="yes" realtime="yes">C:\Users\Demo Windows\Desktop\Wazuh_FIM_Monitor</directories>

  <!-- Frequency that syscheck is executed default every 12 hours -->
  <frequency>43200</frequency>
```

◆ Step 2: Create Test Folder

On agent:

Windows: Create a new folder/file inside the newly created folder (to check if it gets detected in the Wazuh dashboard or not)

1. `cd C:\Users\<username>\Desktop\Wazuh_FIM_Monitor`
2. `echo. > checking_for_fim.txt`

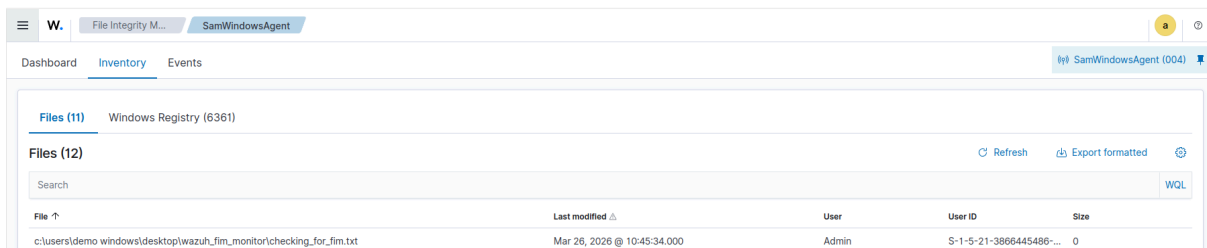
◆ Step 3: Restart Wazuh Agent

Windows:

```
net stop wazuh
net start wazuh
```

◆ Step 4: Check the Wazuh dashboard for an alert

Three horizontal lines → Endpoint security → Threat Hunting → click on Inventory → Refresh (after creating new file under monitored folder)



File	Last modified	User	User ID	Size
c:\users\demo\windows\desktop\wazuh_fim_monitor\checking_for_fim.txt	Mar 26, 2026 @ 10:45:34.000	Admin	S-1-5-21-3866445486-...	0

c:\users\demo windows\desktop\wazuh_fim_monitor\checking_for_fim.txt

Details

Last analysis

Mar 26, 2026 @ 10:45:47.000

User ID

S-1-5-21-3866445486-37083109-4241214515-1001

SHA1

da39a3ee5e6b4b0d3255bfef95601890afd80709

Last modified

Mar 26, 2026 @ 10:45:34.000

Size

0

SHA256

e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855

User

Admin

MD5

d41d8cd98f00b204e9800998ecf8427e

Permissions

Recent events

Search

DQL

Feb 1, 2026 @ 10:34:39.877 → now

Refresh

Add filter

1 hit

Feb 1, 2026 @ 10:34:39.877 - Mar 26, 2026 @ 10:51:58.246

Time	Action	Description	Level	Rule ID
Mar 26, 2026 @ 10:45:48.334	added	File added to the system.	5	554

Table

JSON

Rule

t _index	wazuh-alerts-4.x-2026.03.26
t agent.id	004
t agent.ip	192.168.198.120
t agent.name	SamWindowsAgent

Administrator: Windows PowerShell

```
PS C:\WINDOWS\system32> cd ../../
PS C:\> cd 'C:\Users\Demo Windows\Desktop\'
PS C:\Users\Demo Windows\Desktop> cd .\Wazuh_FIM_Monitor\
PS C:\Users\Demo Windows\Desktop\Wazuh_FIM_Monitor> dir

    Directory: C:\Users\Demo Windows\Desktop\Wazuh_FIM_Monitor

Mode                LastWriteTime         Length Name
----                -
-a- - - -          26/03/2026      03:45           0 Checking_for_FIM.txt

PS C:\Users\Demo Windows\Desktop\Wazuh_FIM_Monitor> Get-FileHash .\Checking_for_FIM.txt -Algorithm md5

Algorithm      Hash                                          Path
-----
MD5            D41D8CD98F00B204E9800998ECF8427E          C:\Users\Demo Windows\Deskt

PS C:\Users\Demo Windows\Desktop\Wazuh_FIM_Monitor> 
```